



Training Data Influences Modularity in Neural Networks

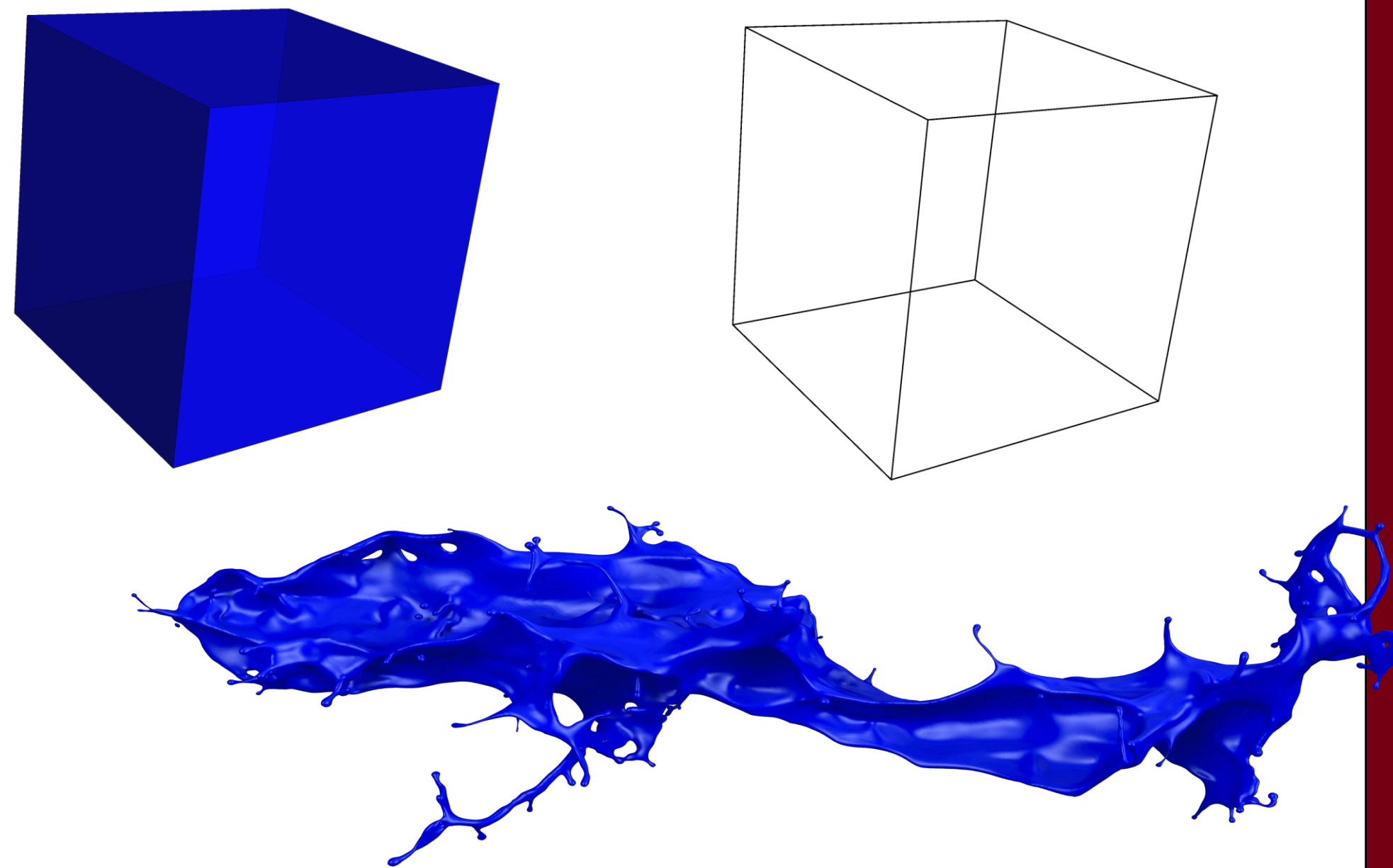


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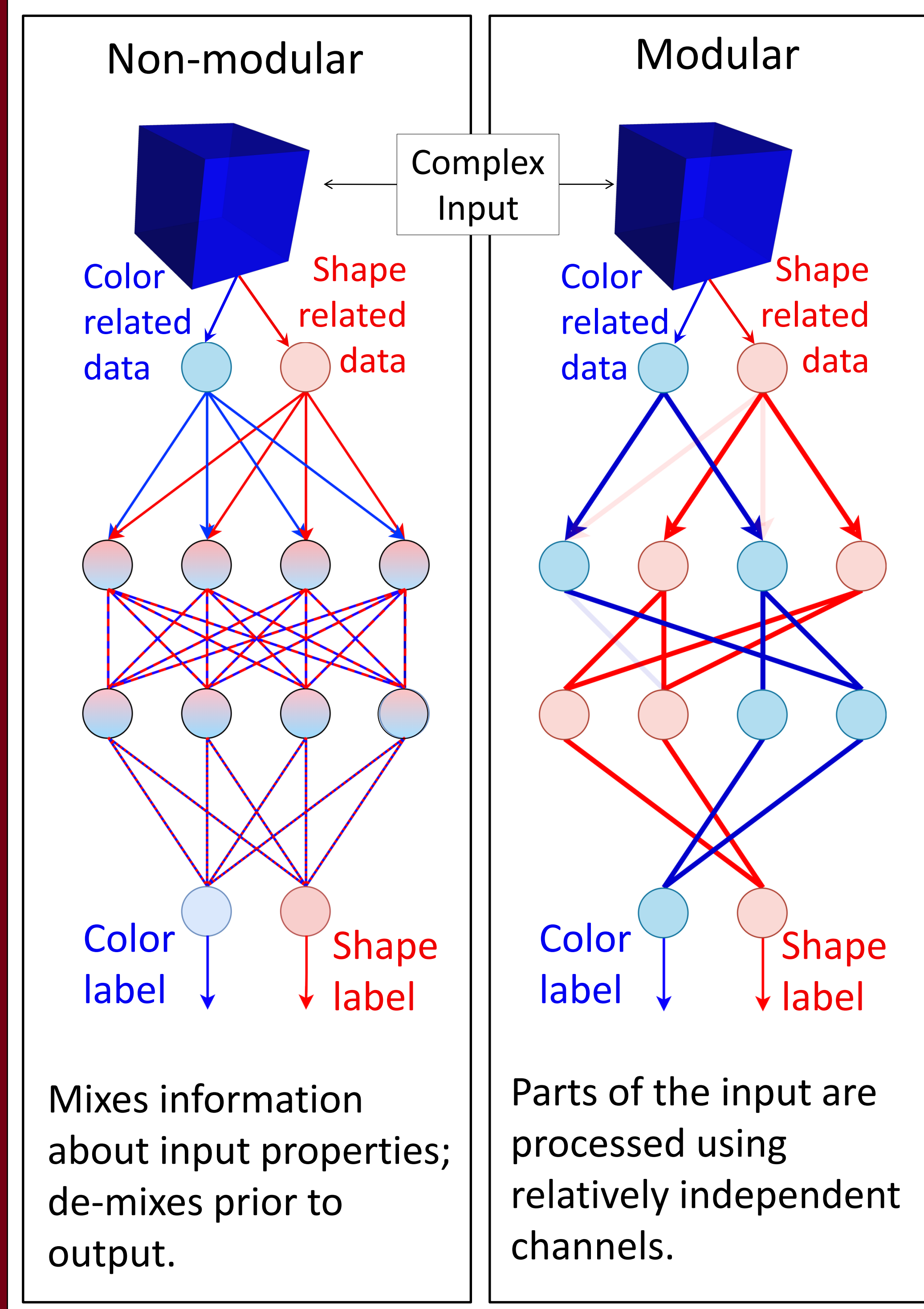
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Many Real World Tasks are Modular

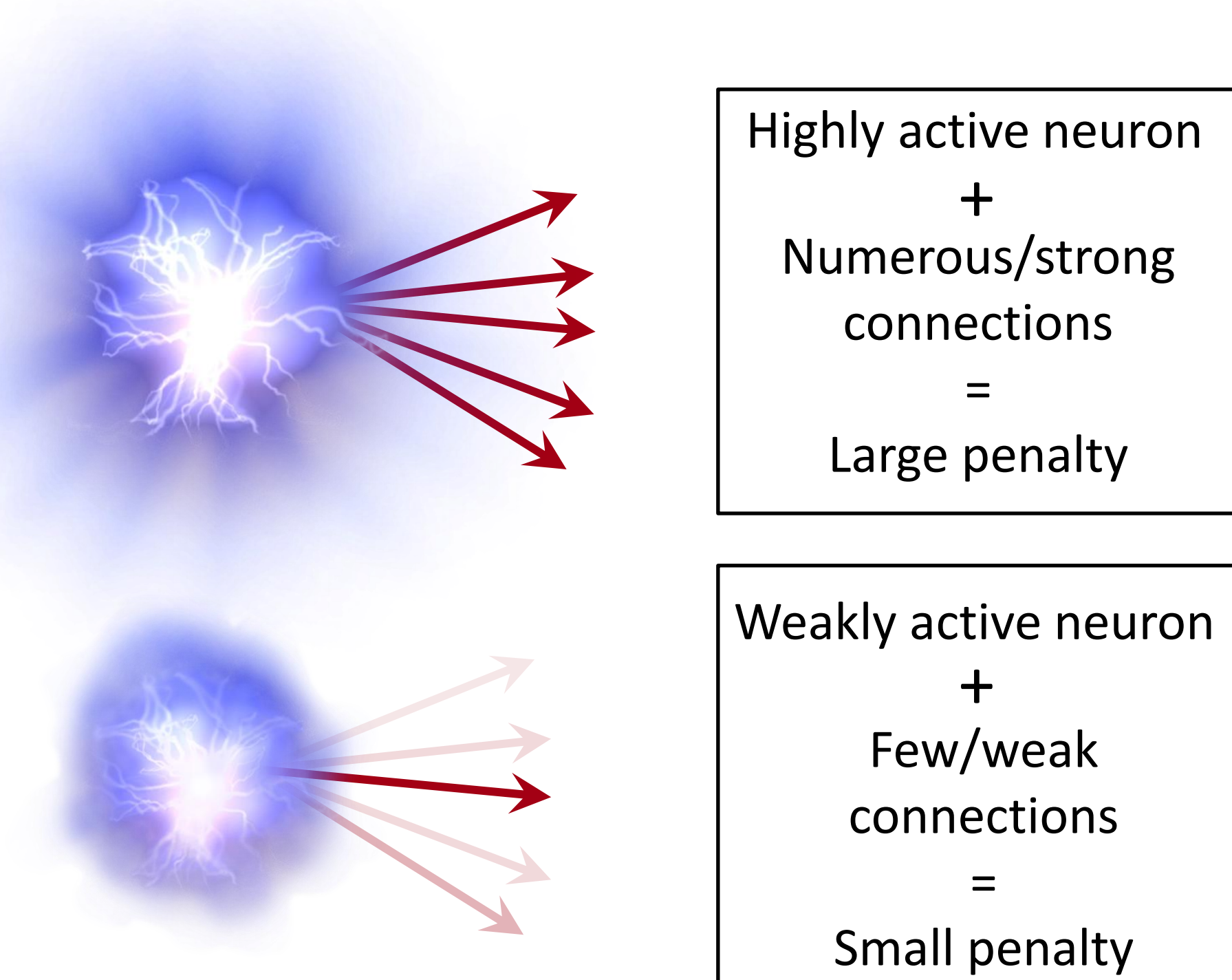


The outputs of modular tasks are determined by separate properties of the input data, which can vary independently of each other.

Neural Networks Can Adopt Modular or Non-Modular Solutions



Nudging Networks Towards Modular Solutions



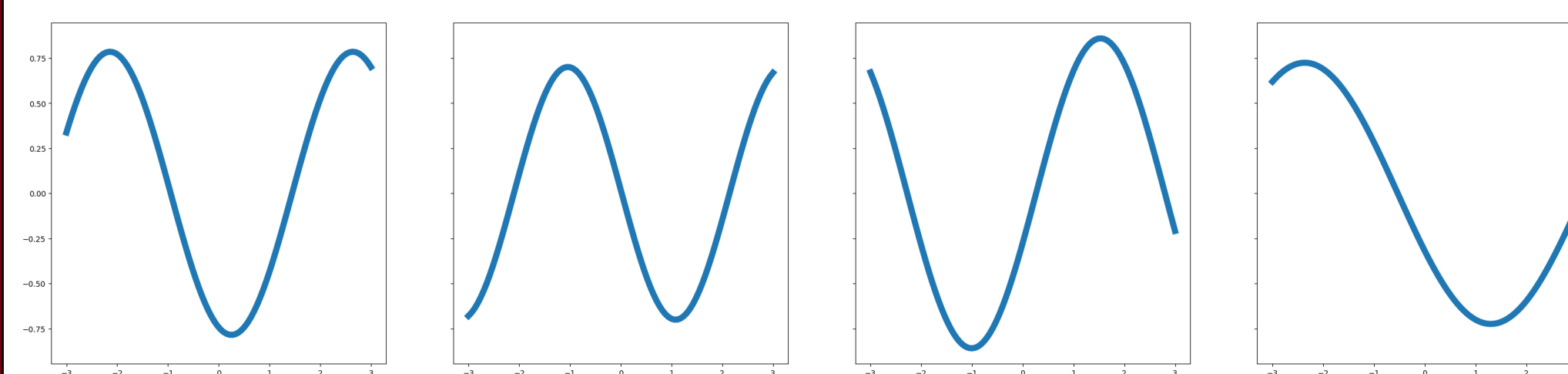
We penalized solutions using highly active and strongly connected neurons, imitating the effects of activity-dependent neural noise, which has been hypothesized to be a driver of modularity.

Training and Measuring Modularity

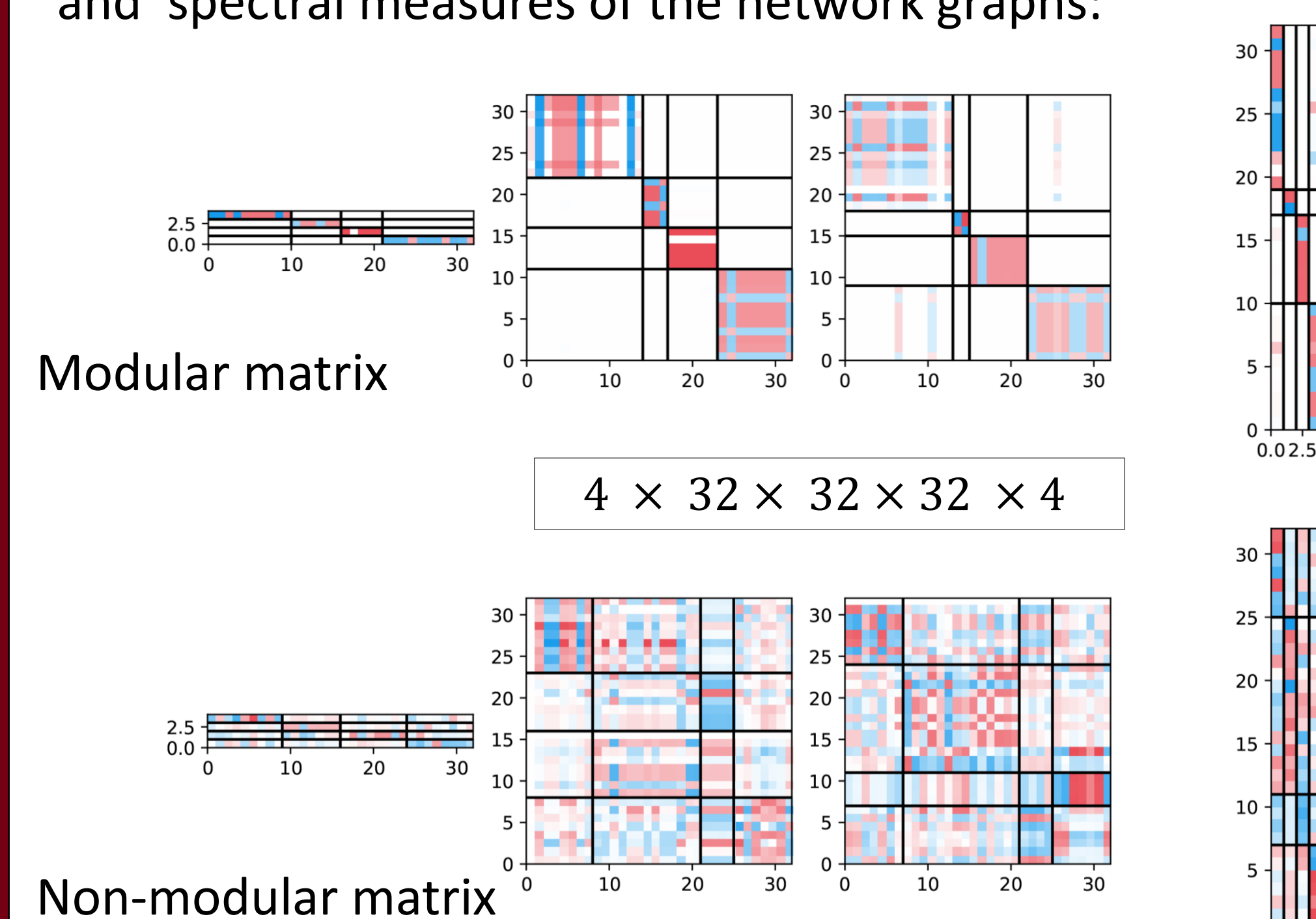
We trained neural networks equipped with the regularizer above on a simple modular regression task:

$$F(x_1, x_2, x_3, x_4) = (f_1(x_1), f_2(x_2), f_3(x_3), f_4(x_4)),$$

where $f_i(x_i)$ has form $A \sin(B x_i + C)$ and $x_i \in [-2, 2]$.

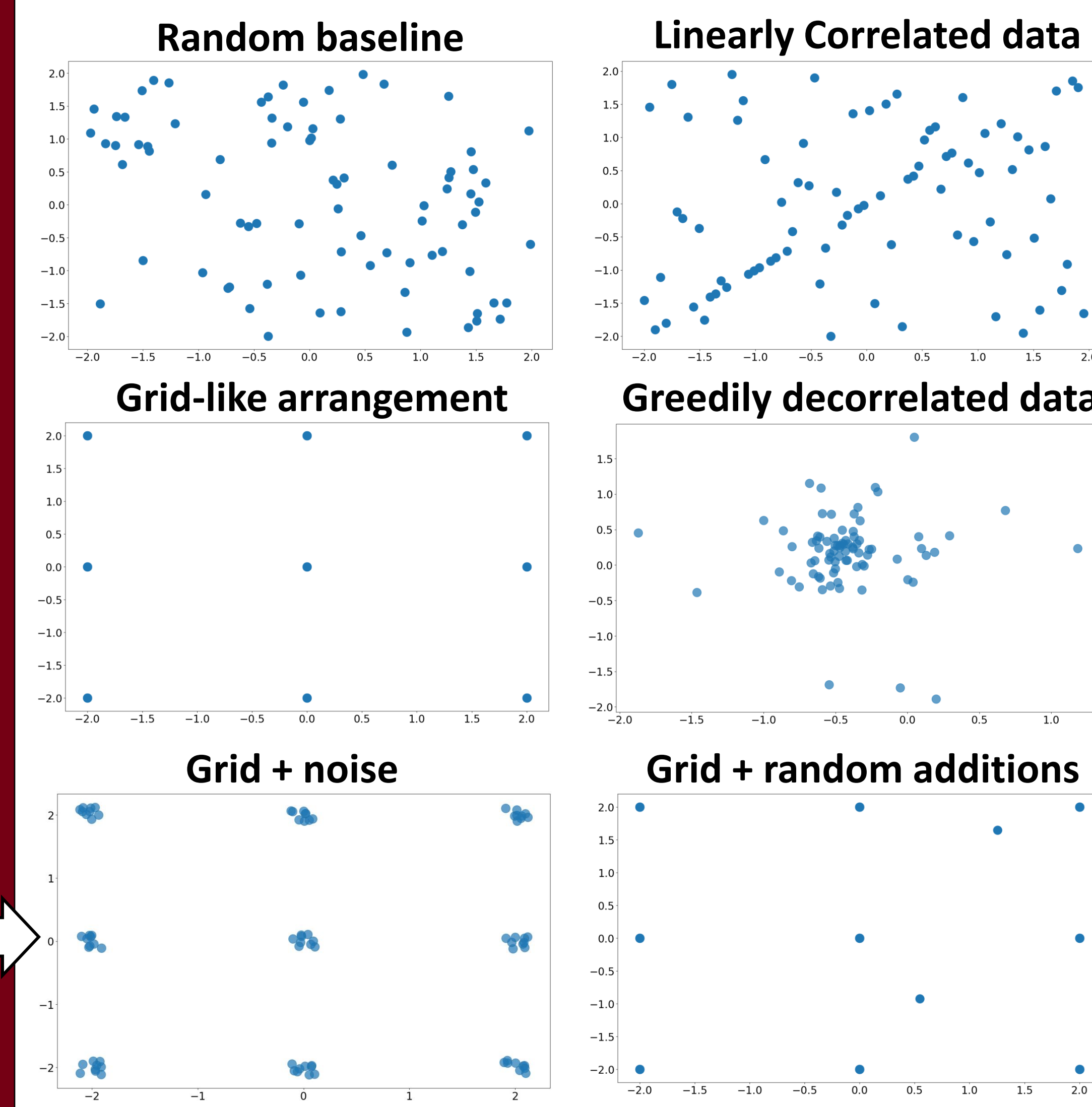


We assessed modularity through connectivity matrices and spectral measures of the network graphs:



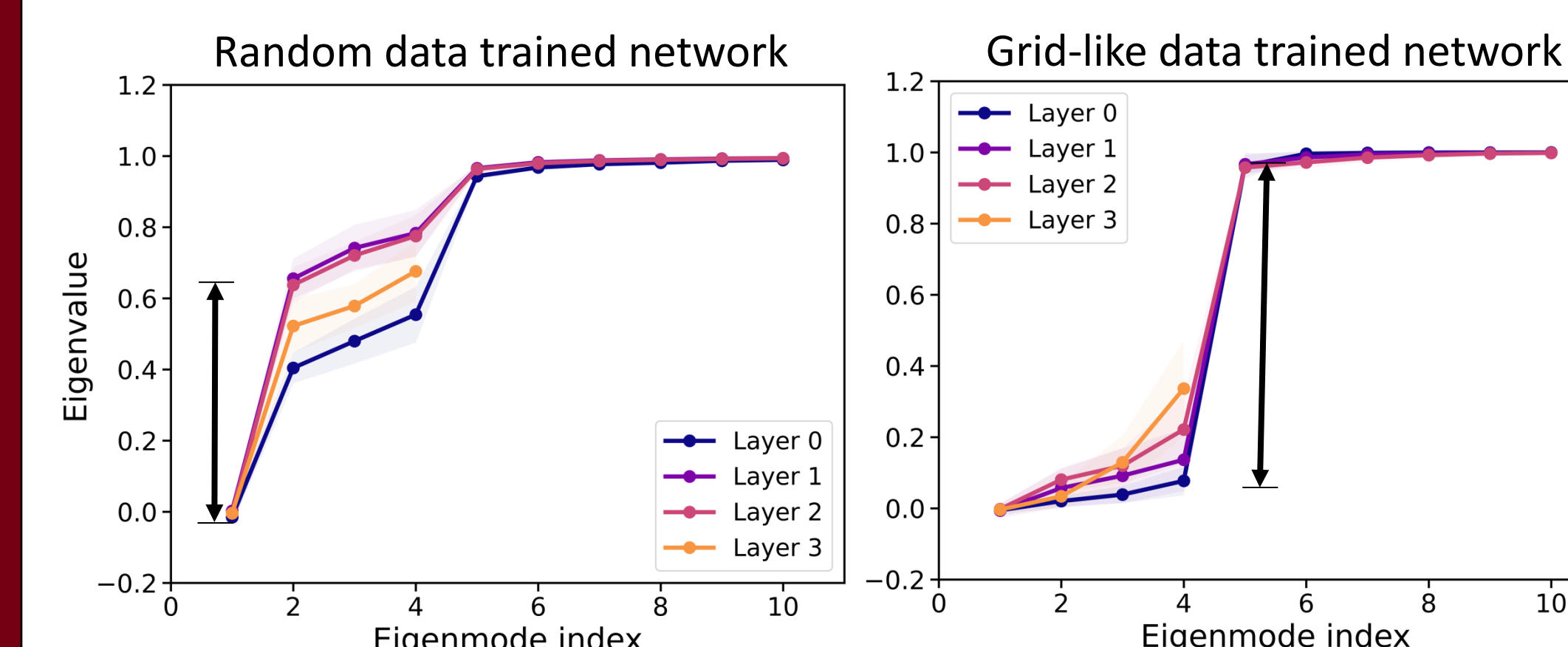
How Does Training Data Affect Modularity?

We created training datasets with various characteristics (2D projections of 4D datasets shown):



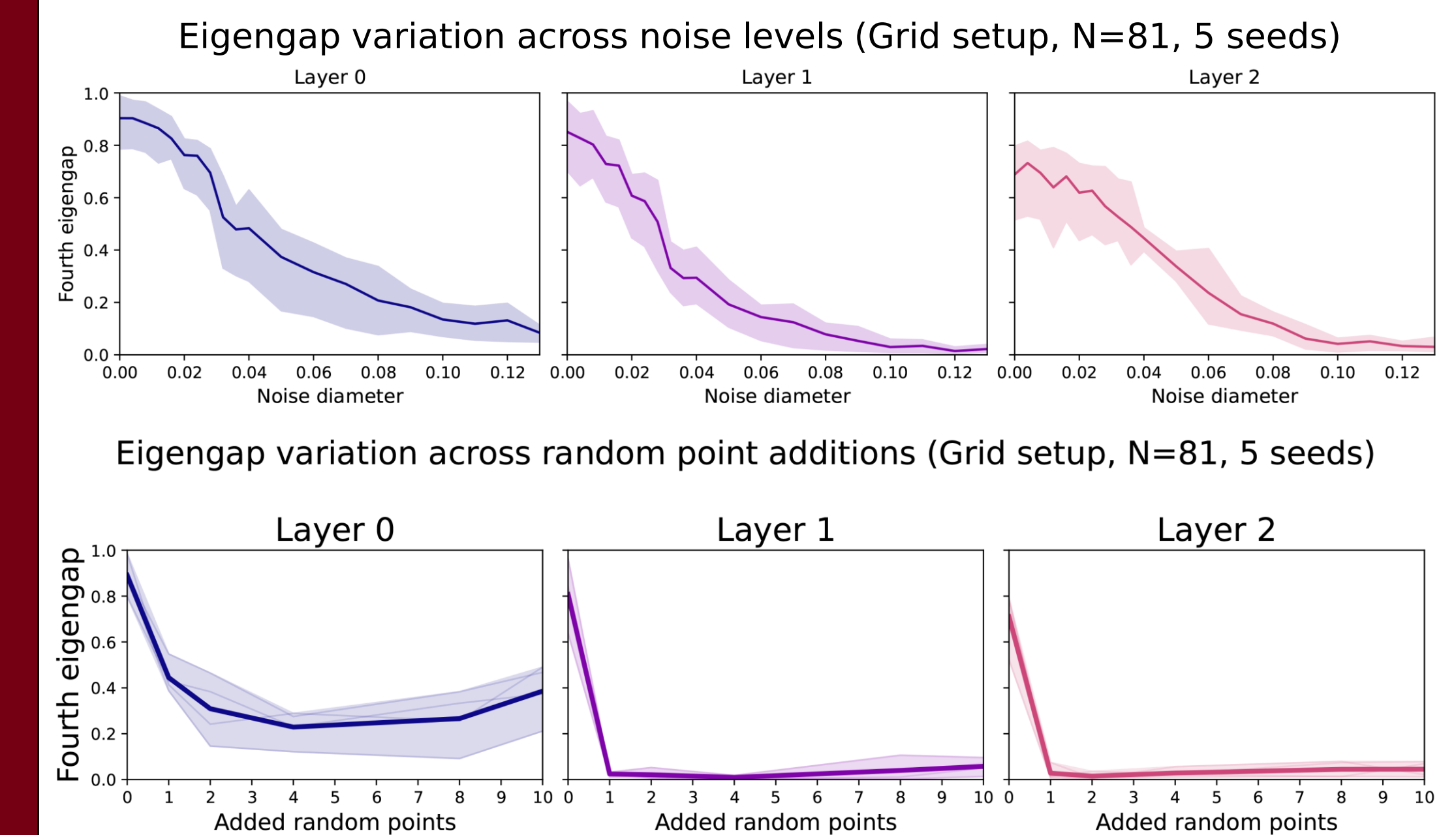
Grid-like data induces much stronger modularity than random data at small sample sizes

Training datasets had 81 samples.
Mean eigenvalues across 10 seeds. Shaded area = ± 1 std.

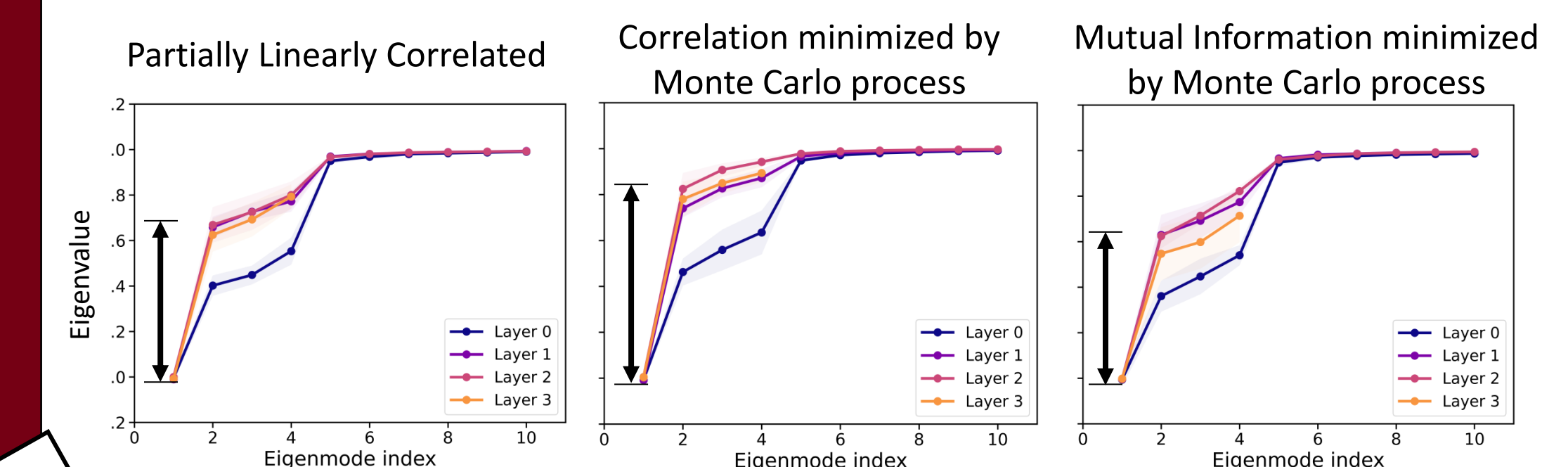


Modularity can be characterized via the eigenspectra of the graph Laplacian matrices of the connectivities. Clusters are detected as eigenvalues near 0. Eigenvalues close to 1 correspond to trivial structures. The number of clusters is identified by finding the largest difference between neighboring eigenvalues (double arrows).

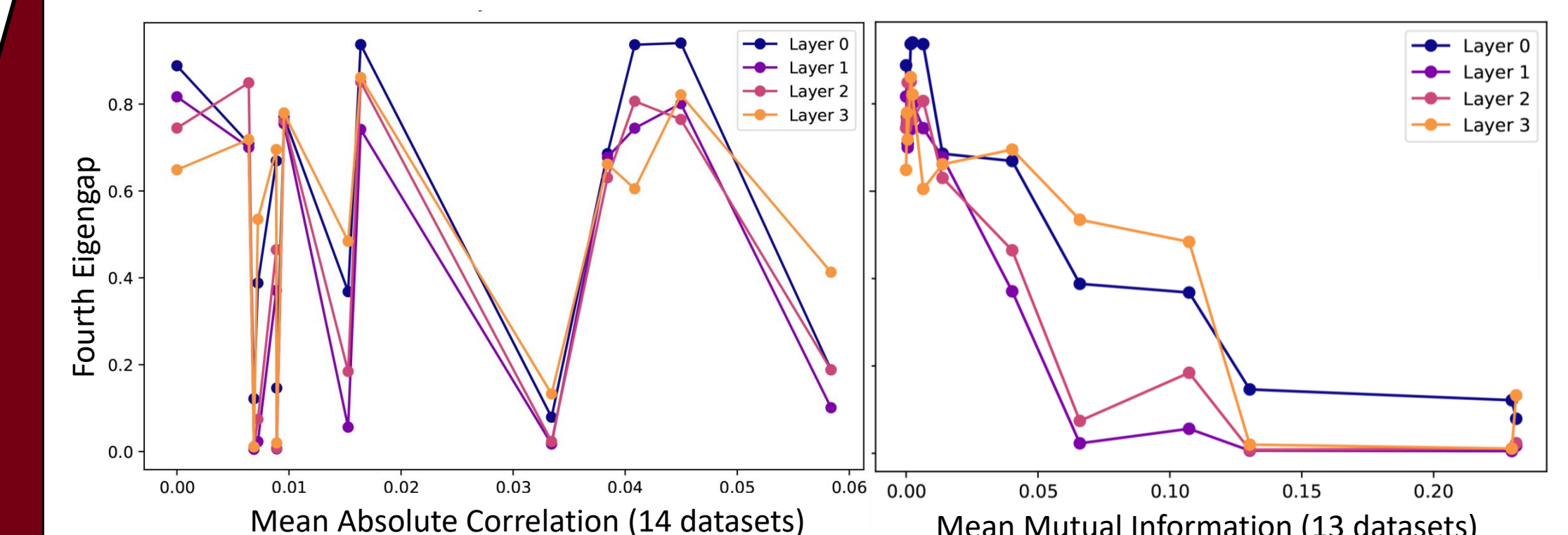
Grid effectiveness survives some noise and point removals, but not additions of few random points



Correlation inhibits modularity, but greedily reducing it alone is not enough



Characterizing Datasets



Mean absolute correlation between input properties is a poor predictor of which datasets induce modularity (left). Mutual information tracks this more closely in our tested datasets (right). However, it is possible to construct datasets with zero MI that still fail to induce modularity.

Conclusions and Future Directions

- Training data greatly affects how weighted-activity regularized networks modularize.
- Repeated grid-like contrasts among input properties strongly induce modularity, even at small sample sizes.
- Statistical dependence measures alone don't seem to fully explain this effect.
- Future work could precisely characterize this contrast structure and exploit it for training data curation.

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